Seat No.:_____

AHU ID No._____ Name_____ Major_____ Grade_____

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Stony Brook Institute at Anhui University

2023 Spring Semester

Applied Calculus /// Final Exam (Paper A)

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	Total score
Score					
Grader					

I. Blank Fillings (5 items, 3 points each)

Score	

1. Let $(a,b,c)=1$, then $((a+b)\times(b+c))\cdot(c+a)=$_____.

2. Exchange the order of the integral $\int_0^2 dy\int_{y^2}^{2y} f(x,y)\,dx=$_____.

3. Let S be the surface of a tetrahedron with vertices $O(0,0,0)$, $A(1,0,0)$, $B(0,1,0)$, $C(0,0,1)$ with the outward orientation. Then $\iint_S (x+1)\,dydz+ydzdx+(xy+z)\,dxdy$ = _____.

4. The solid region Ω is bounded by the surface $z=\sqrt{x^2+y^2}$ and the plane $z=1$. Then the volume Ω is _____.

5. If $\dfrac{(x+ay)\,dx+ydy}{(x+y)^2}$ is the total differential of a function, then $a=$ _____.

II. Choices (5 items, 3 points each)

Score	

6. If the position relationship of two lines $\dfrac{x-1}{1}=\dfrac{y+1}{2}=\dfrac{z-1}{\lambda}$, $\dfrac{x+1}{1}=\dfrac{y-1}{1}=\dfrac{z}{1}$ is intersecting, then λ is ( )

A. 1. B. $\dfrac{3}{4}$. C. $-\dfrac{5}{4}$. D. $\dfrac{5}{4}$.

7. Let $L: y=x^2$ $x\in[0,1]$, $I_1=\int_L k(y-x^2)\,ds$, $I_2=\int_L (y-x)\,ds$, k is a constant, then ( )

A. $I_1 < I_2$. B. $I_1 > I_2$. C. $I_1 = I_2$. D. uncertain .

8. Suppose the upper hemispherical surface is $S: x^2 + y^2 + z^2 = R^2 \left(z \geq 0\right)$, then the unique integral which value is not zero is ( )

A. $\iint\limits_S x^2 dS$. B. $\iint\limits_S y dS$. C. $\iint\limits_S x dS$. D. $\iint\limits_S y^3 dS$.

9. If $\sum\limits_{n=0}^{\infty} a_n \left(x+1\right)^n$ is convergent at the point $x=1$, then at $x=0$ it is ( )

A. divergent. B. conditionally convergent.
C. absolutely convergent. D. uncertain.

10. If the series $\sum\limits_{n=0}^{\infty} a_n^2$ is convergent, then $\sum\limits_{n=0}^{\infty} \left(-1\right)^{n-1} a_n$ is ( )

A. absolutely convergent. B. conditionally convergent.
C. divergent. D. either convergent or divergent.

III. Calculations (6items, 9 points each)

Score	

11. Evaluate $\iiint\limits_{\Omega} z dx dy dz$, in which $\Omega = \left\{\left(x, y, z\right) \middle| x^2 + y^2 + z^2 \leq z\right\}$.

12. If the curve $l: \begin{cases} x^2 + y^2 + z^2 = 6 \\ x + y + z = 0 \end{cases}$, find the tangent line equation and the normal plane equation of the curve at the point $P_0\left(1, -2, 1\right)$.

13. Suppose the curve L is $\frac{x^2}{3}+y^2=1$ with the counter-clockwise. Calculate $I=\oint_L \frac{xdy-ydx}{x^2+3y^2}$.

14. Find $\iint_S 2x^3dydz+2y^3dzdx+3\left(z^2-1\right)dxdy$, where S is the surface $z=1-x^2-y^2\left(z\geq 0\right)$ with the upward orientation.

15. Calculate the sum function $S(x)$ of the power series $\sum_{n=0}^{\infty}(2n+1)x^n$.

16. Represent the function $f(x)=\frac{1}{3-x}$ as the power series of $(x-1)$.

IV. Proof Problem (2 items, 8 points each)

Score	

17. Prove the integral $\int_L \left(x^2 - yz\right)dx + \left(y^2 - zx\right)dy + \left(z^2 - xy\right)dz$ is independent of the path.

18. Let $\lim_{x \to 0} \frac{f(x)}{x} = 0$, prove that the series $\sum_{n=1}^{\infty} f\left(\frac{1}{n^2}\right)$ is absolutely convergent.